

# VX-400

CONTROL VALVE / Flow Control

An installation and application reference for the VX-400 control valve. Use it to frame piping orientation, actuator setup, instrument checks, and the project-specific configuration review.



| PLATFORM ROLE                    | REFERENCE STATUS             | DOCUMENT USE                   |
|----------------------------------|------------------------------|--------------------------------|
| Modulating flow control platform | Illustrative project concept | Early engineering conversation |

TECHNICAL SPECIFICATION

| PARAMETER      | REFERENCE ENVELOPE   |
|----------------|----------------------|
| SIZE RANGE     | DN25-DN300           |
| PRESSURE CLASS | PN16-PN100           |
| TEMPERATURE    | -40 C to 400 C       |
| ACTUATION      | Pneumatic / Electric |
| LEAKAGE CLASS  | VI                   |

The values above are a reference envelope for concept discussions. Confirm duty, materials, performance, and compliance requirements against the approved project specification before procurement or manufacture.

# VX-400 / ENGINEERING NOTES

## CONTROL VALVE / CONFIGURATION FRAMEWORK

The platform is configured around the operating envelope rather than selected from a single catalogue point. Use the prompts below to structure the application review.

## DESIGN INTENT AND INTERFACES

| ENGINEERING LENS              | WHAT TO RESOLVE                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Control intent</b>         | Provide repeatable modulation across the defined flow range with an actuator and position feedback strategy suited to the service. |
| <b>Installation attention</b> | Flow direction, pipe support, access to trim and actuator, upstream conditioning, and instrument air or power quality.             |
| <b>Release boundary</b>       | Valve size, trim, actuator, seals, fail position, materials, and leakage basis require project approval before manufacture.        |

## CONFIGURATION REVIEW

| ENGINEERING LENS      | WHAT TO RESOLVE                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Valve station</b>  | Keep the body supported by the piping arrangement and leave a practical path for actuator adjustment, inspection, and removal.       |
| <b>Actuator setup</b> | Verify travel direction, fail position, signal range, position feedback, and any accessories against the approved control narrative. |
| <b>Commissioning</b>  | Check free travel, zero and span, loop response, isolation, and the as-left setpoints before introducing process conditions.         |

## ENGINEERING HANDOFF CHECKLIST

|           |                                                        |
|-----------|--------------------------------------------------------|
| <b>01</b> | Line class, flow direction, and process data confirmed |
| <b>02</b> | Trim, actuator, fail action, and signal basis selected |
| <b>03</b> | Piping support and service access checked              |
| <b>04</b> | Loop test, setpoints, and as-left record prepared      |

|                       |                                                                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>REFERENCE NOTE</b> | This document is a polished portfolio reference, not a certified data sheet. Exact performance, materials, interfaces, testing, and applicable requirements remain project-specific and require written confirmation. |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|